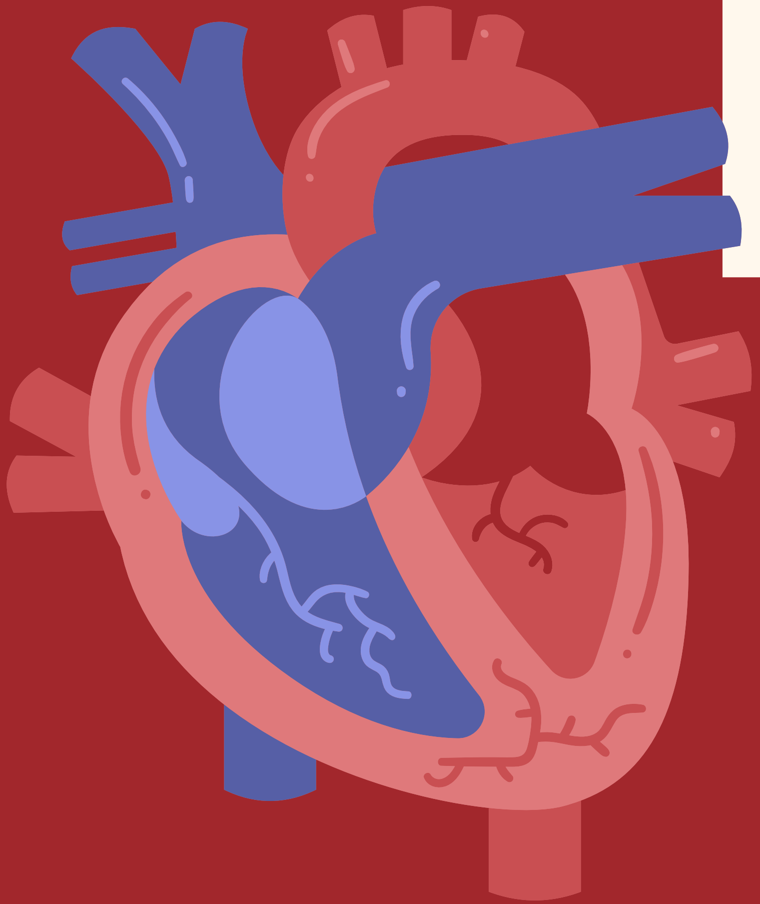
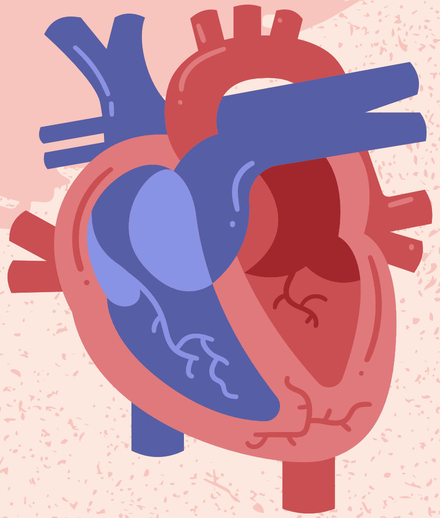
STRUCTURE AND PROPERTIES OF HAEMOGLOBIN

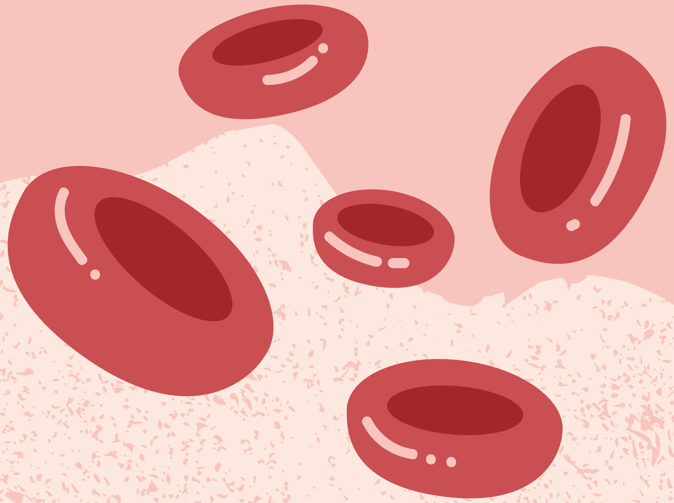


Submitted by
RASITHA K

INTRODUCTION



Haemoglobin is an iron containing and oxygen carrying red pigmet

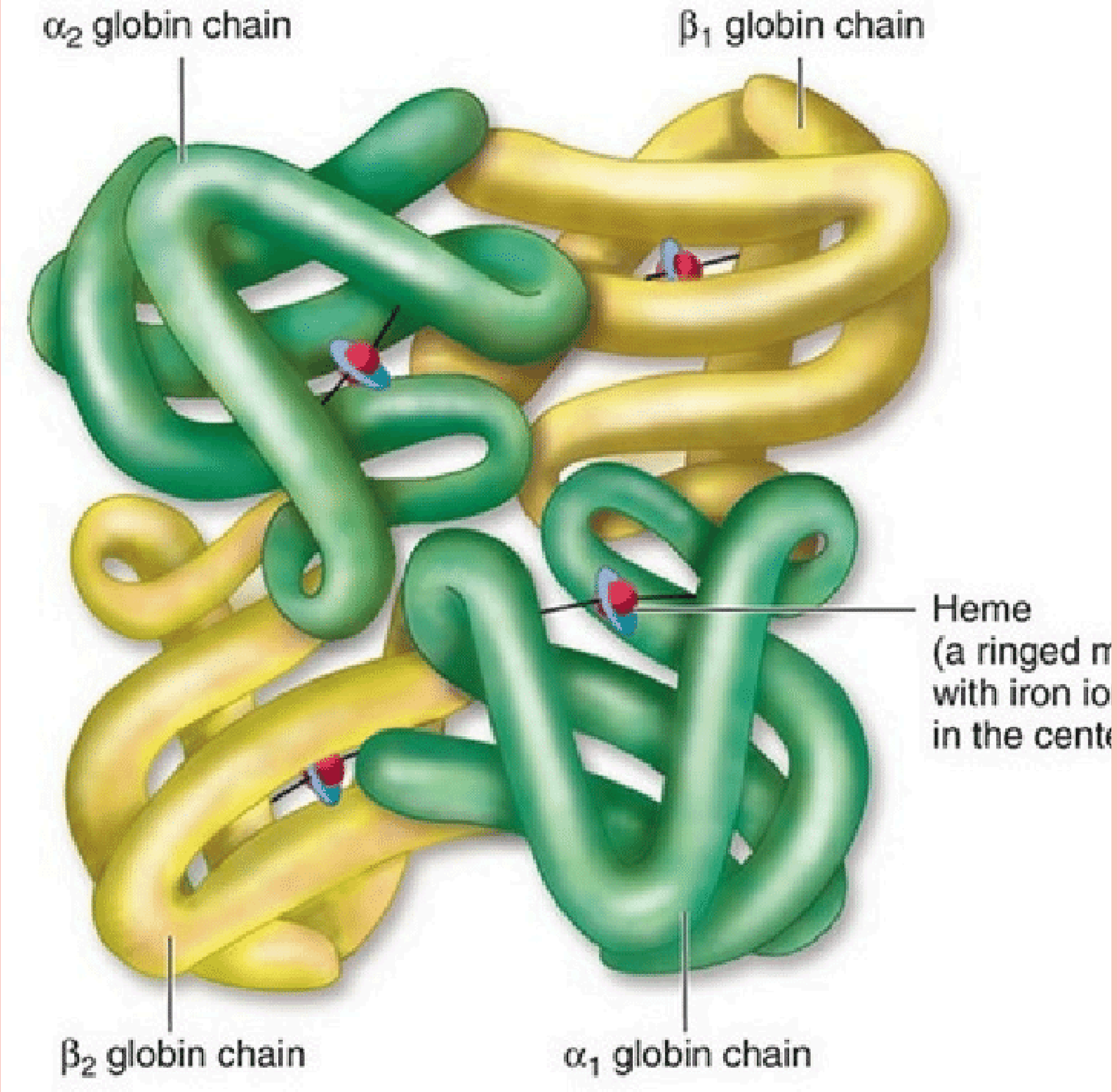


Essential for oxygen delivery to tissues and organs

BASIC STRUCTURE

- Hemoglobin is a tetramer composed of four subunits.
- Each subunit consists of a globin protein and a heme group.
- HbA consists of one globin molecule and four haem molecule

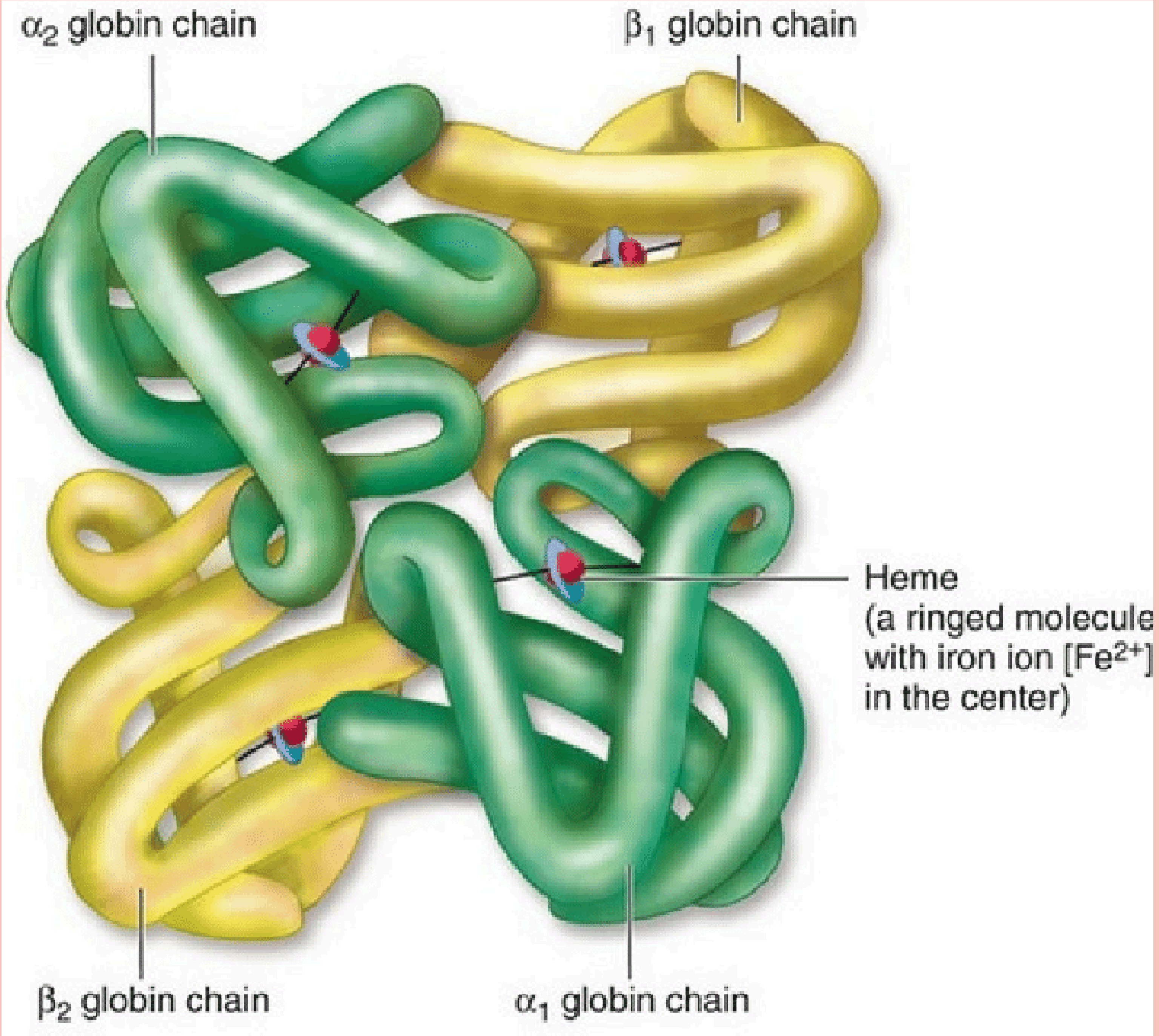
VISUAL DIAGRAM



SUBUNIT COMPOSITION

GLOBIN

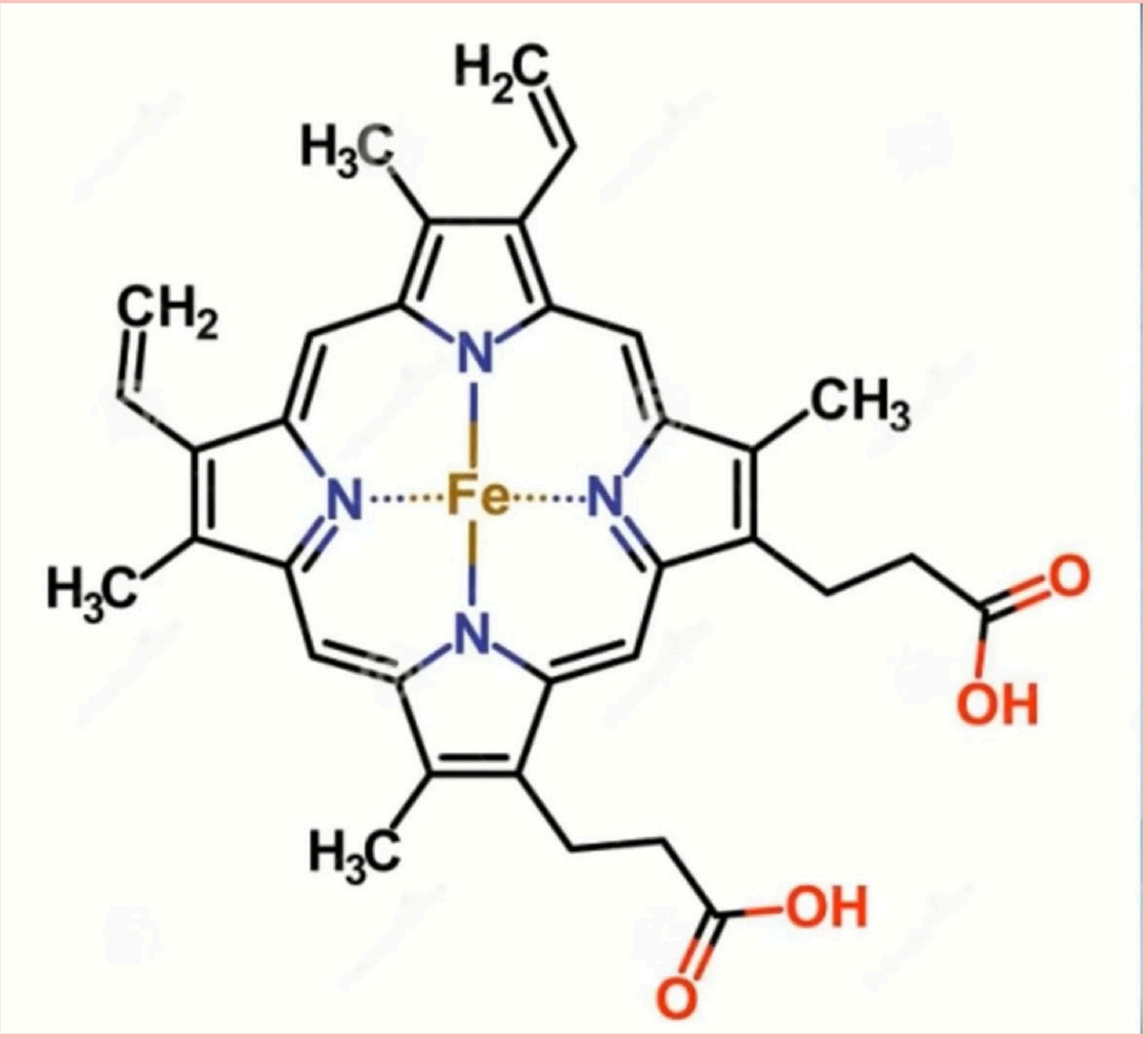
- Tetrameric protein formed of four polypeptide chains
- Two alpha chains and two beta chains



SUBUNIT COMPOSITION

Haem

- Each heme group contains an iron (Fe) atom at its center.
- Haem molecule is an iron porphyrrin complex
- porphyrrin unit is a tetrapyrrole ring



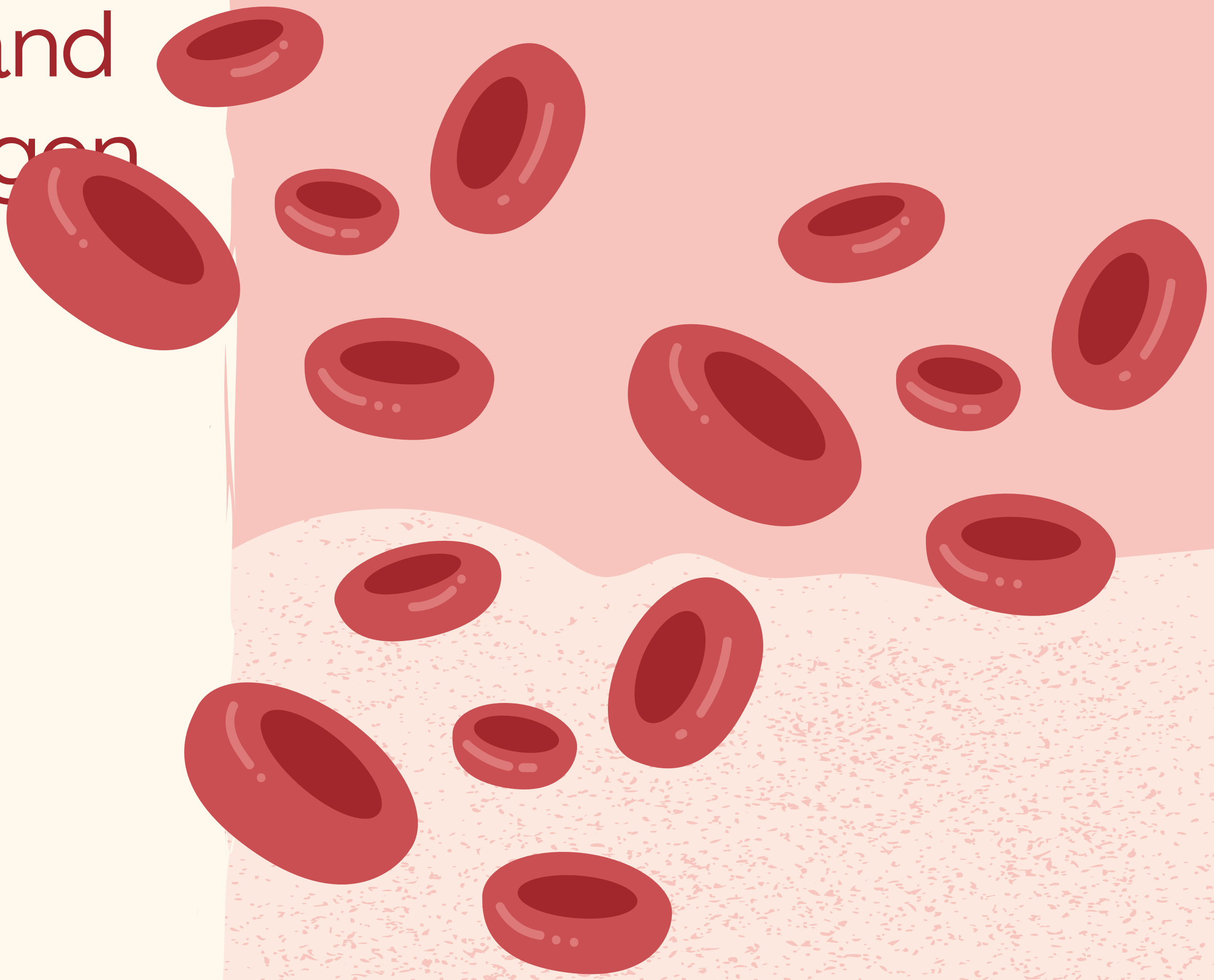
PROPERTIES

1

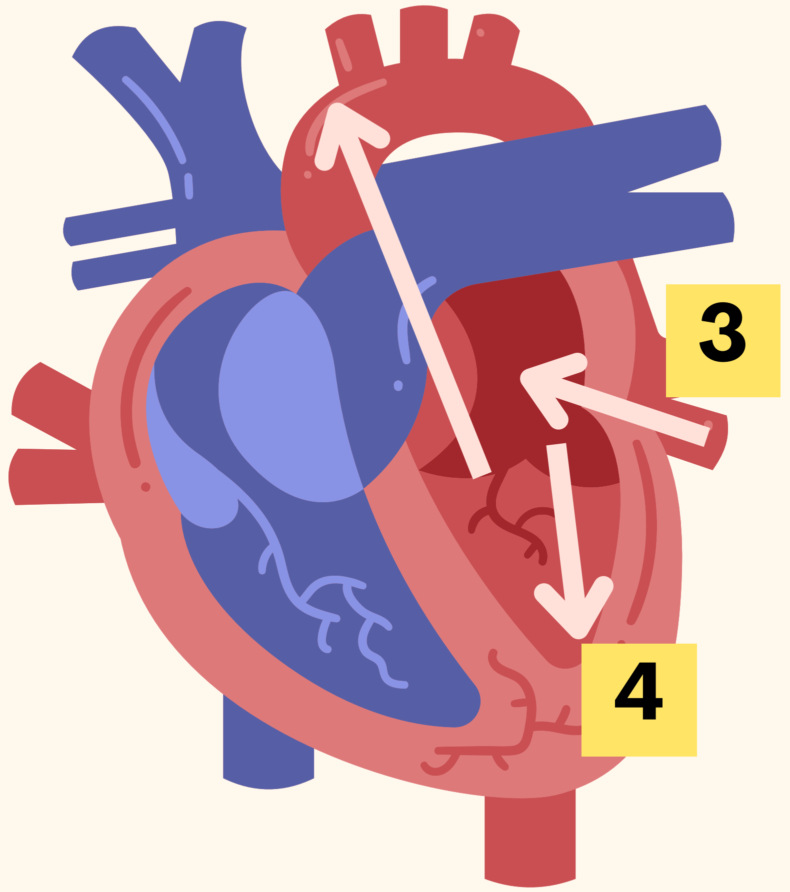
- Haemoglobin is its high affinity for O_2 and great ability to reversibly bind with oxygen
-

2

Affinity for oxygen depends on temperature,pH,p$O_2$and pCO_2 of the blood,and the concentration of 2,3-diphosphoglycerate in the blood



TYPES OF HAEMOGLOBIN



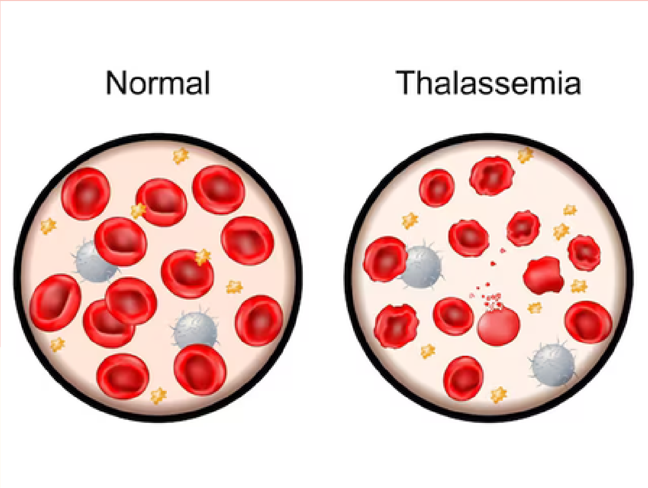
1. HbA2-Globin part is formed of two alpha chains and two delta chains

2. Haemoglobin content of the blood of a normal human adult consists of about 97%HbA,2%HbA2,And 1%HbF

ABNORMAL HAEMOGLOBINS

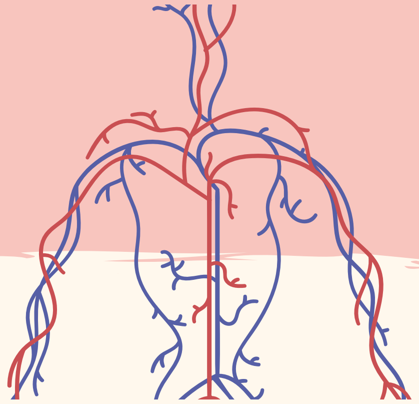
HbS

Abnormal haemoglobin found in sickle-cell anaemia patients.



HbC

- Abnormal haemoglobin found in thalassemia patients.
- Normal glutamic acid in the 5^th position of the beta chain of HbA is replaced by lysine.



Haemoglobin D,HaemoglobinE

HbD-Replacement of beta 121 glutamic acid by glutamine
HbE-Replacement of beta 26 glutamic acid by lysine.

LESSON CHECK: THE HEART

Can you explain how blood flows through the heart with a partner, even without the diagram of the heart?

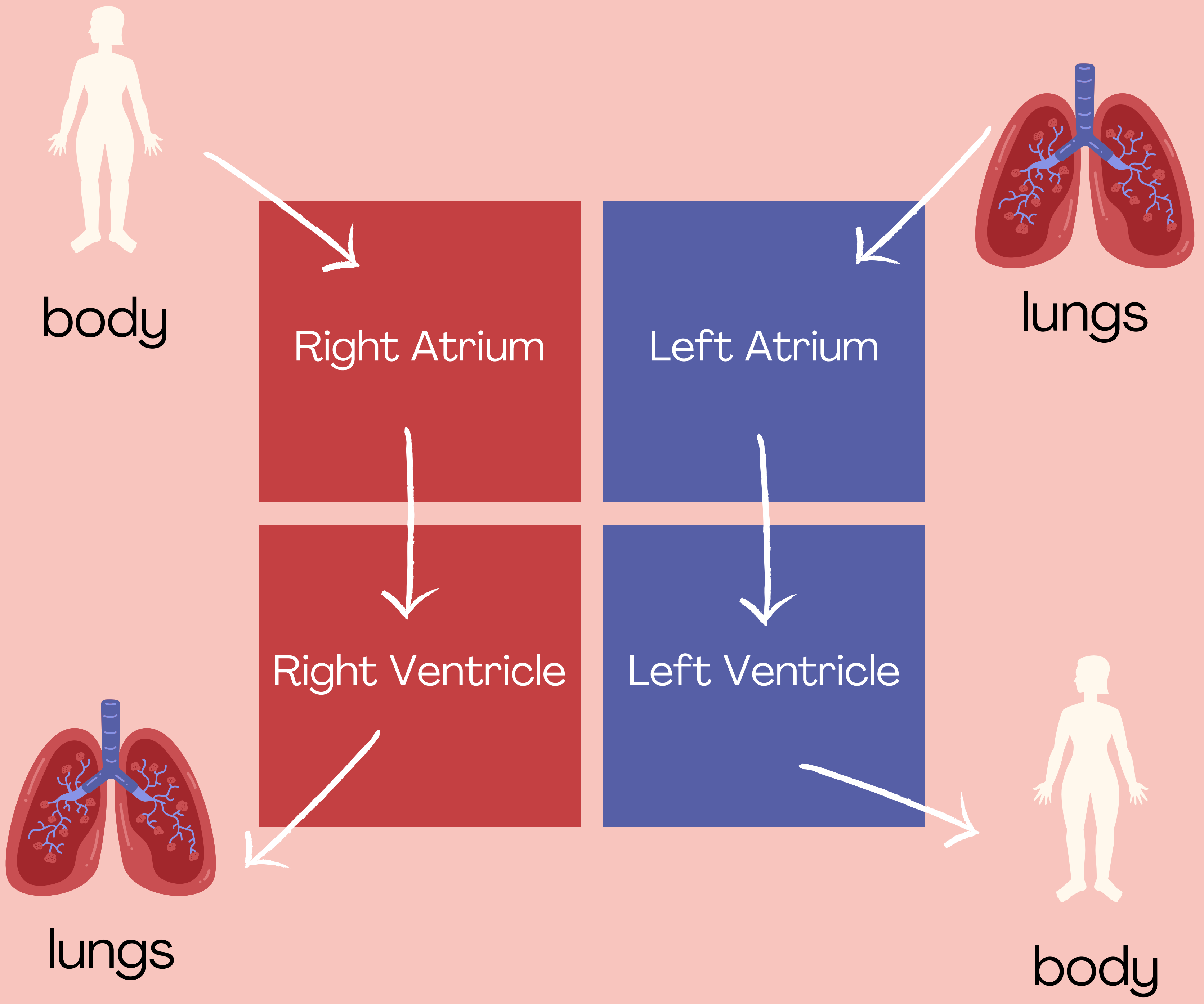
Try explaining the process to your peer without referring to your notes.

SAMPLE ANSWER

LESSON CHECK: THE HEART

Can you explain how blood flows through the heart with a partner, even without the diagram of the heart?

Shown is a sample diagram of how blood flows through the heart.



BLOOD

It is a special **fluid** primarily contained within the blood vessels.

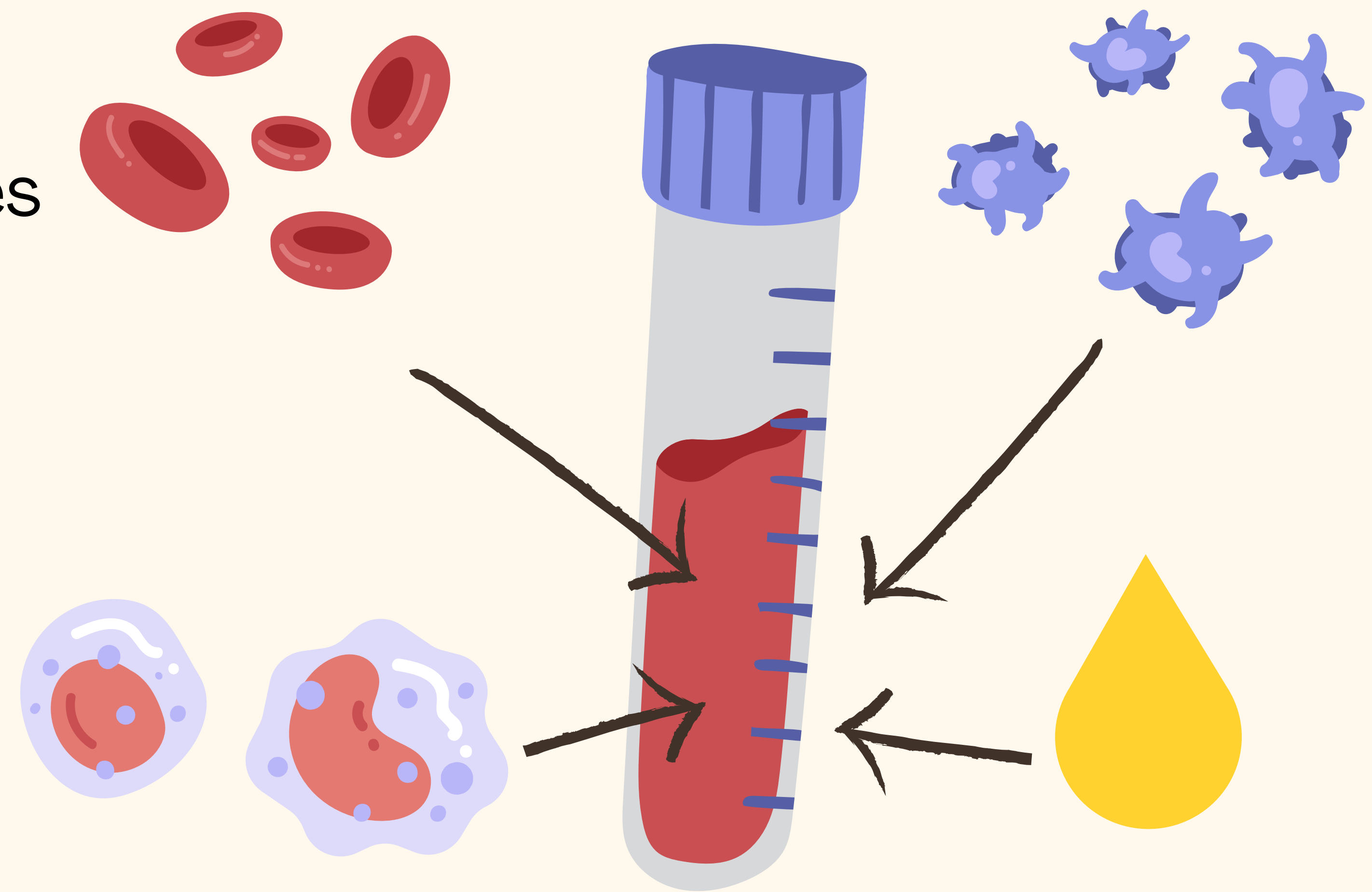
It has **four main components**—red blood cells, white blood cells, platelets and plasma.

BLOOD

Our blood has four main components:

Red Blood Cells

carry oxygen, nutrients and wastes



Platelets

gather at the site of injury and help the clotting process

White Blood Cells

fight diseases and protect the body from infection

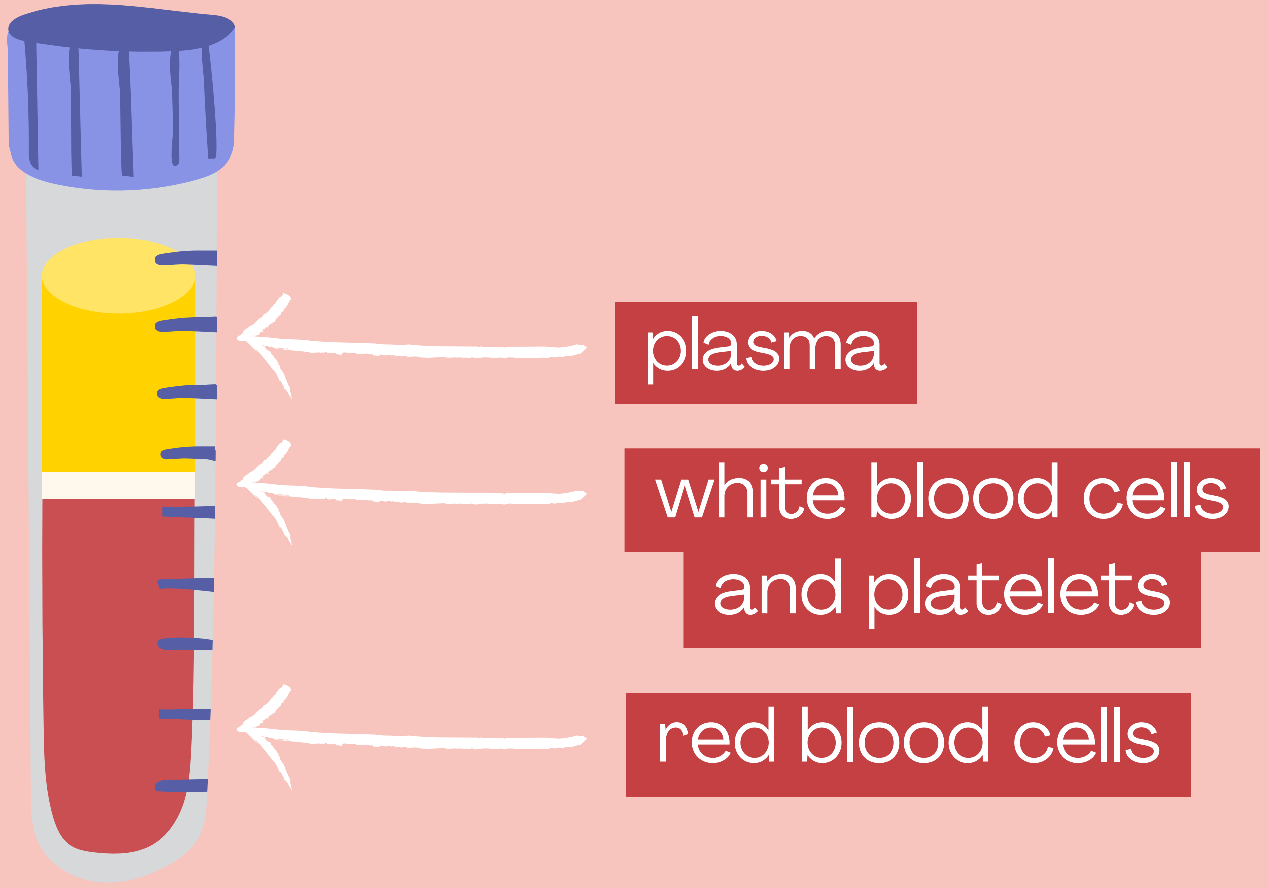
Plasma

straw-coloured liquid where the other components float in

BLOOD

When someone has a blood test, a small amount of blood is taken and kept in a tube for testing.

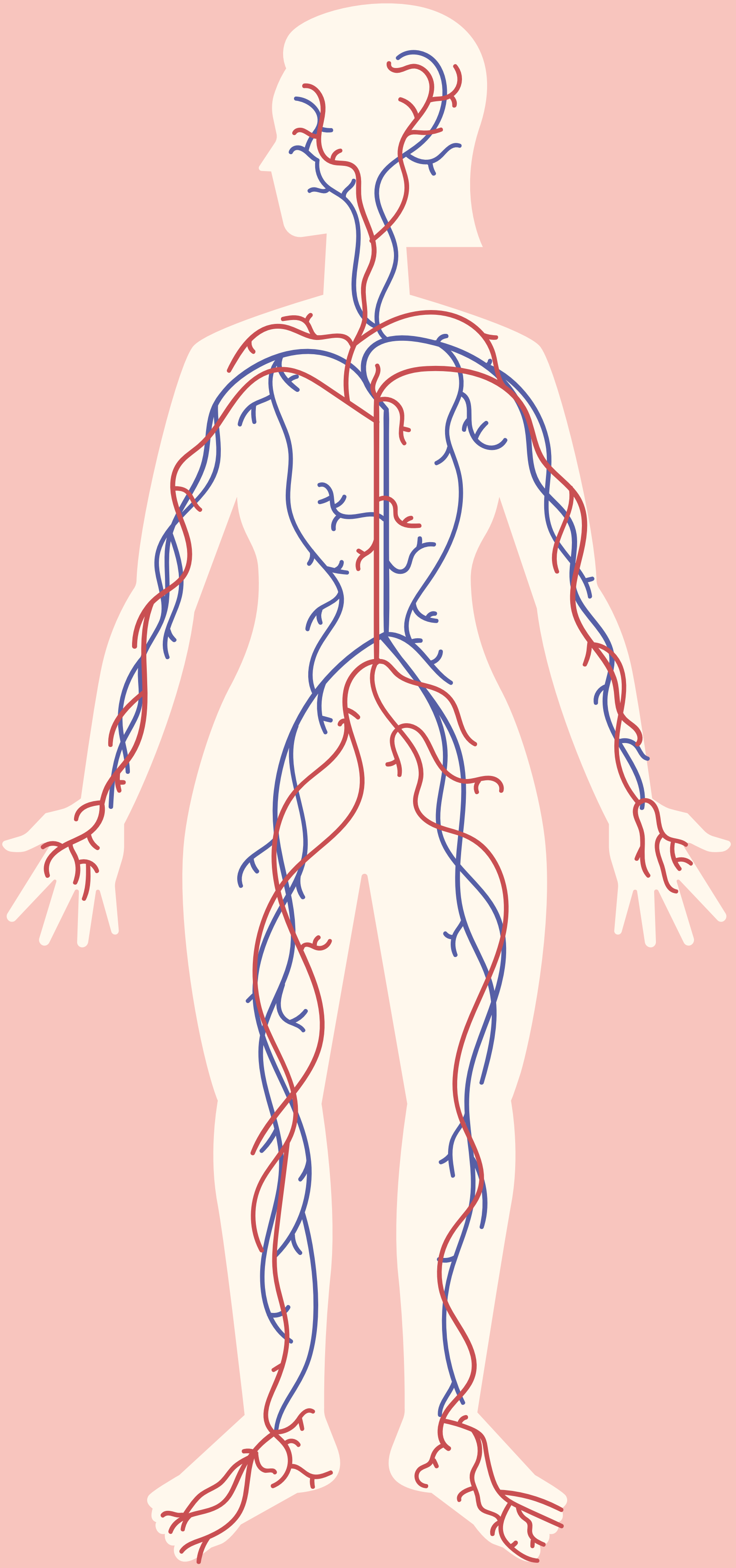
To closely examine it, blood undergoes component separation. One common method involves spinning it at very high speeds in a centrifuge.



Blood is divided into its components, with the heaviest parts at the bottom.

FUNCTIONS OF HAEMOGLOBIN

- Transport of resoiratory gases
- Serve as a buffer system and regulates blood pH
- undergoes disintegration and forms the bile pigments



BLOOD VESSELS

Let's take a closer look at the three types of blood vessels:

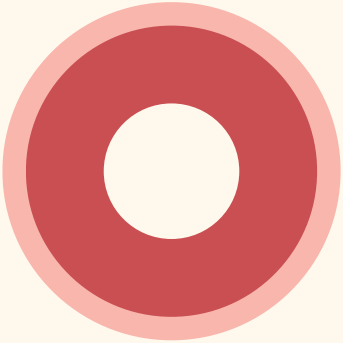
Artery It has the **thickest wall** of all three, allowing it to withstand the high pressure created by the heart.

Capillary It has the **thinnest wall** to allow substances such as oxygen and sugars to pass through its wall—into or out of the blood.

Vein It is **less muscular and stretchy** than an artery, so blood moves through it with low pressure. It also has a special valve that helps blood go only one way.

BLOOD VESSELS

Each type of blood vessel has a unique role in the circulatory system.

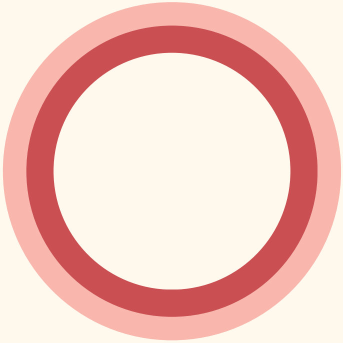


Artery

Carries blood **away from the heart**

Capillary

Assists in the **exchange of substances** between the blood and tissues



Vein

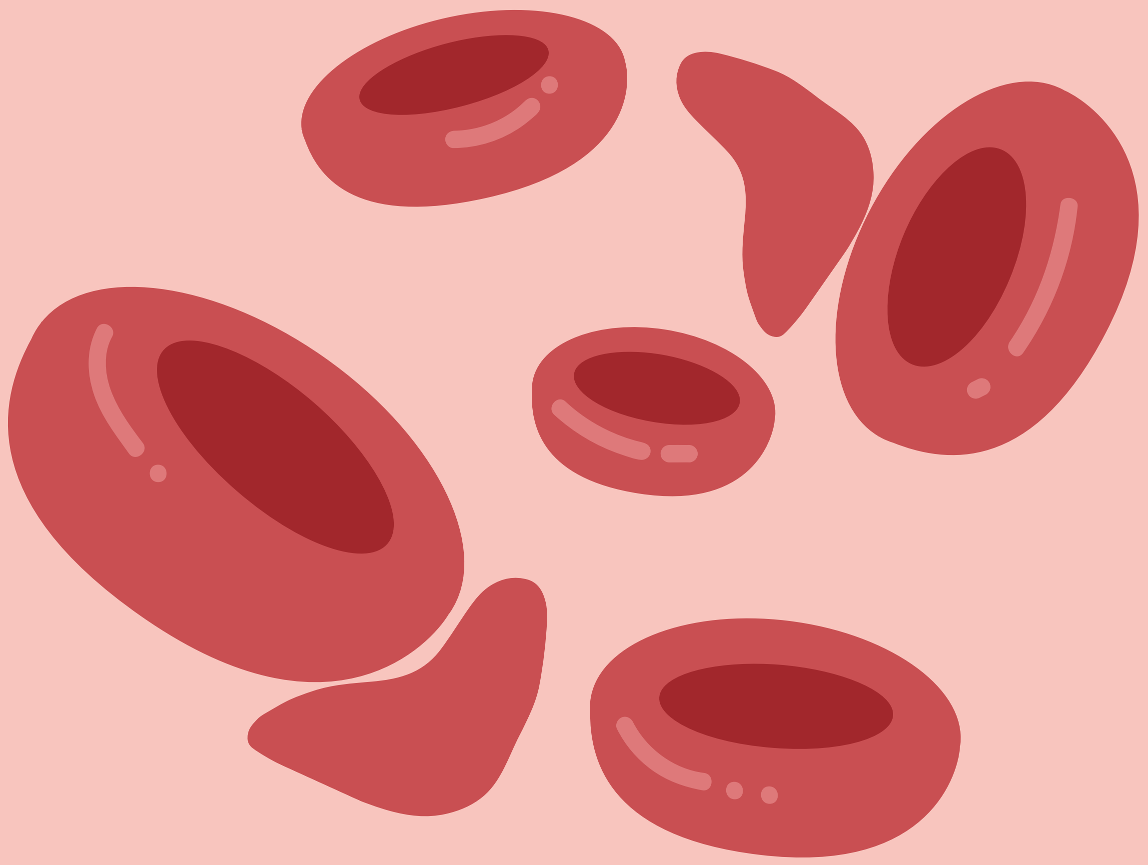
Carries blood back **towards the heart**

TIP Remember 'A' for 'artery' and 'away' from the heart!

LESSON CHECK: SICKLE CELL DISEASE

Sickle cell disease (SCD) is an inherited blood disorder. People with SCD have red blood cells that become **hard and sticky**, forming a **C-shaped blood cell** instead of the healthy disc-shaped one.

Considering what you've learnt about blood and blood vessels, how does this shape affect blood flow in the circulatory system?



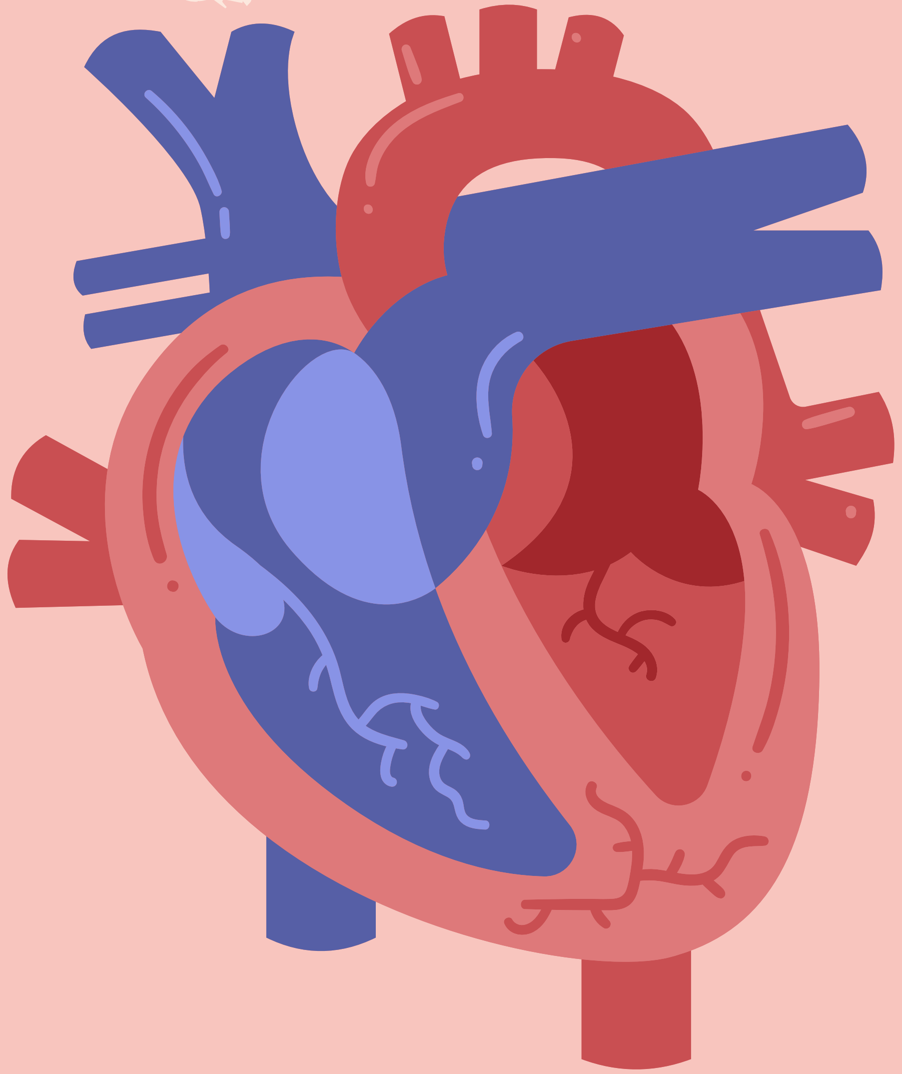
SAMPLE ANSWER

LESSON CHECK: SICKLE CELL DISEASE

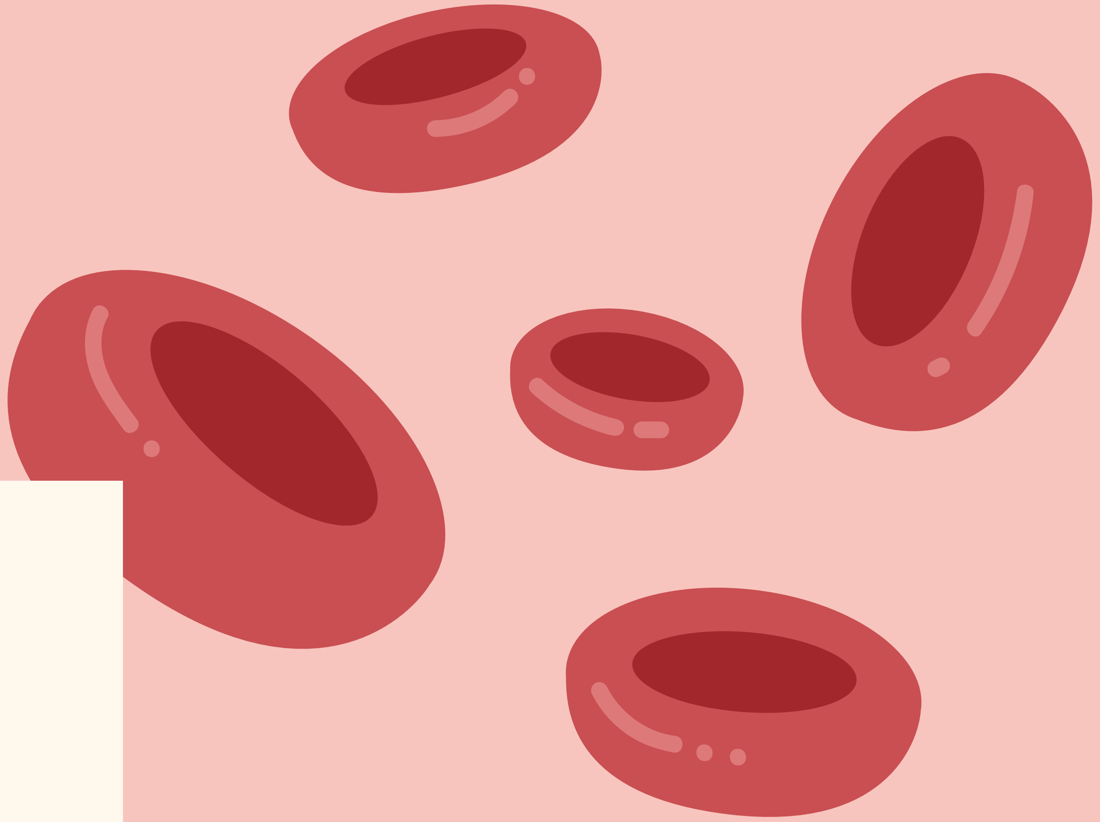
Considering what you've learnt about blood and blood vessels, how does this shape affect blood flow in the circulatory system?

The irregular shape of the sickle-shaped blood cell makes it easy to clog and block the blood flow.

This can cause pain and other serious infections.



THANK YOU.



PRESS THESE KEYS WHILE ON PRESENT MODE!

- **B** for blur
- **C** for confetti
- **D** for a drumroll
- **M** for mic drop
- **O** for bubbles
- **Q** for quiet
- **U** for unveil
- **0-9** Any number from 0-9 for a timer